

# MANAN RASTOGI

+91 9548594935 | [mananrastogi2k8.210@gmail.com](mailto:mananrastogi2k8.210@gmail.com) | [linkedin.com/in/manan-rastogi](https://linkedin.com/in/manan-rastogi) | [github.com/xmananrastogi](https://github.com/xmananrastogi)  
ECE undergrad (VIT) · IIT Madras DS&A · Applied AI · Full-Stack · Computer Vision

## EDUCATION

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| <b>Vellore Institute of Technology (VIT) — Vellore, India</b><br><i>B.Tech in ECE (Biomedical Specialization)   CGPA: 7.53</i> | <b>2024 – Present</b> |
| <b>IIT Madras — Online, India</b><br><i>BS in Data Science &amp; Applications</i>                                              | <b>2025 – Present</b> |
| <b>Delhi Public School — Bareilly, India</b><br><i>Class XII — 80%   Class X — 80%</i>                                         | <b>Graduated 2022</b> |

## EXPERIENCE

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| <b>Summer Intern — Retail Automation</b><br><i>Indian Oil Corporation</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>May 2026 – Jun 2026</b><br><i>Bareilly, India</i> |
| <ul style="list-style-type: none"><li>- Built <b>ComplaintGuard</b> — a Streamlit + Python app that replaced the team's manual Excel SLA review. A batch that used to take hours now runs in about <b>30 seconds</b>.</li><li>- Reads IOCL's vendor complaint exports, works out whether each ticket was Early, On Time, or late against its SLA (24h or 48h), and totals up <b>INR 1,000/day</b> penalties — results go out as a formatted Excel sheet.</li><li>- Also catches auto-closed tickets and flags equipment visited twice within 30 days — things the old VBA macro couldn't handle (and which broke entirely on Mac).</li></ul> |                                                      |

## PROJECTS

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| <b>WoundTrack AI   Python, OpenCV, Flask, SciPy, SQLite</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <a href="#">Live Demo</a>   <a href="#">GitHub</a> |
| <ul style="list-style-type: none"><li>- Tracks <b>441 cells</b> across <b>800+ frames</b> in under 10 minutes on a regular laptop — <b>28% more accurate</b> than the lab's ImageJ workflow, which needed manual annotation per frame.</li><li>- Detection uses LoG blob-finding then a LAP gap-closing tracker with Kalman-filter predictions, so trajectories survive occlusions without fragmenting mid-sequence.</li><li>- Researchers upload an experiment on the Flask dashboard and get kinetic curves, MSD breakdowns, and one-click exports — no more copy-pasting between spreadsheets.</li></ul>                           |                                                    |
| <b>VITalize AI   Next.js 15, TypeScript, MongoDB, LLM APIs, Tesseract OCR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">Live Demo</a>                          |
| <ul style="list-style-type: none"><li>- Academic platform for VIT students — <b>9,103 past papers</b> searchable by subject, AI-generated solutions (LLaMA 3.1 + OCR) with KaTeX math rendering, <b>2,439 courses</b> and <b>2,587 calendar events</b> tracked.</li><li>- The FFCS solver finds clash-free schedules across <b>2,400+ combinations</b> in under a second using backtracking with constraint pruning for faculty, time gaps, and slot clashes.</li><li>- Chrome extension (Manifest V3) pulls live data from VIT's VTOP portal directly into the platform. Deployed serverless on Vercel with MongoDB Atlas.</li></ul> |                                                    |
| <b>3D Engineering Portfolio   React, TypeScript, Three.js (R3F), Tailwind CSS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">Live Demo</a>   <a href="#">GitHub</a> |
| <ul style="list-style-type: none"><li>- Personal site with interactive Three.js 3D scenes, lazy-loaded bundle splitting, and sub-1s initial load on mobile — built as a real engineering project, not just a design exercise.</li><li>- Full WCAG pass: keyboard navigation, ARIA labels, skip-to-content, reduced-motion fallbacks. Lighthouse: <b>100 SEO, 97 accessibility, 100 best practices</b>.</li></ul>                                                                                                                                                                                                                      |                                                    |

## CERTIFICATIONS

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| <b>Designing Cisco Security Infrastructure (SDSI)   Credly Badge</b> | <b>Jul 2026</b> |
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## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript  
**AI/ML & CV:** OpenCV, NumPy, SciPy, LLM APIs, Tesseract OCR  
**Web:** React, Next.js, Three.js, Flask, Streamlit, Tailwind CSS, REST APIs, JWT  
**Databases:** MongoDB, SQLite, SQL  
**Tools:** Git, GitHub, Vercel, VS Code